Count Regression

Homework Environmental Statistics 2

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This weeks exercises will practice the content of the lecture and tutorials and *extend* them in different ways:

- step-wise model selection;
- BIC as criterion of model quality;
- plot univariate relationship from a multivariate model ("conditional plot").

Please try to **switch off** all automatic reporting of model selection steps or package warnings, which both only swamp the PDF you generate without providing useful information.

Exercise 1 (20 points): Plant species richness on Galapagos islands

Galapagos is a very remote archipelago. Its unique fauna and flora have attracted biologists, ecologists and evolutionary biologists for hundreds of years. The data set `gala` (in **faraway**) gives the total and endemic number of plant species for each of the 30 islands of Galapagos, along with some geographic information. What are the key drivers of plant species richness? (Hint: The number of species is typically a power-law function of area *A*: $S = cA^z$. On a log-log-scale, this becomes $\log S = \log c + z \log A$. Using count data, *we automatically log-transform the response*, so we additionally need to use $\log$ area as predictor.)

1. Plot the logarithm of the number of plant species against log(area), elevation and distance to nearest island, separately. (**2 points**)
2. Formulate a GLM for count data (Poisson), using the above three variables as predictors. Add pairwise interactions and, if the plot suggests a non-linear relationship, also quadratic terms. Is this a "good model" (explained deviance, dispersion, number of predictors, etc.)? (**3 points**)
3. Repeat the same plotting, analysis and critique for the number of endemic plant species. (**2 points**)
4. Repeat the species richness analysis using a negative binomial GLM. Does that make a difference to (i) model parameters and their errors, (ii) significances, and (iii) dispersion? (**3 points**)

5. Repeat step 4 for the *endemic* species richness. (**2 points**)
6. Simplify both negative binomial models automatically, using backwards stepwise model selection with BIC as criterion. (**3 points**)
7. Plot both data and the models (with 95% confidence intervals) for the effect of log(area); use the untransformed scale for the response. Make the plot size 5 x 5 inches (not the default 4 x 4). (**3 points**)
8. Interpret the findings, particular the figure from step 7. Pay attention to whether the power-law slopes ("z" in the above equation) are similar or not. (**2 points**)

In [19]:
```
# add other necessary packages

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm
import statsmodels.formula.api as smf
from statsmodels.formula.api import ols
from sklearn.linear_model import LinearRegression, PoissonRegressor
from sklearn.metrics import mean_squared_error
%pip install pingouin
import pingouin as pg
from patsy import dmatrices
import scipy
%pip install mlxtend
from mlxtend.feature_selection import SequentialFeatureSelector

import warnings
```

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Requirement already satisfied: pingouin in .\anaconda3\Lib\site-packages (0.7.0)
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Note: you may need to restart the kernel to use updated packages.
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```

```
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Requirement already satisfied: threadpoolctl>=3.5.0 in .\anaconda3\Lib\site-packages
(from scikit-learn>=1.8.0->mlxtend) (3.5.0)
Note: you may need to restart the kernel to use updated packages.
```

In [20]:

```python
# call dataset
gala = pd.read_csv("gala.csv")

# Reset the index by replacing with name
gala = gala.set_index('Name')
gala.describe()
```

Out[20]:

	Species	Endemics	Area	Elevation	Nearest	Scruz	Adjacent
count	30.000000	30.000000	30.000000	30.000000	30.000000	30.000000	30.000000
mean	85.233333	26.100000	261.708667	368.033333	10.060000	56.976667	261.098333
std	114.633053	27.331111	864.110519	421.604937	14.274636	68.032334	864.518967
min	2.000000	0.000000	0.010000	25.000000	0.200000	0.000000	0.030000
25%	13.000000	7.250000	0.257500	97.750000	0.800000	11.025000	0.520000
50%	42.000000	18.000000	2.590000	192.000000	3.050000	46.650000	2.590000
75%	96.000000	32.250000	59.237500	435.250000	10.025000	81.075000	59.237500
max	444.000000	95.000000	4669.320000	1707.000000	47.400000	290.200000	4669.320000

In [21]:
```
gala.head(10)
```

Out[21]:

Name	Species	Endemics	Area	Elevation	Nearest	Scruz	Adjacent
Baltra	58	23	25.09	346	0.6	0.6	1.84
Bartolome	31	21	1.24	109	0.6	26.3	572.33
Caldwell	3	3	0.21	114	2.8	58.7	0.78
Champion	25	9	0.10	46	1.9	47.4	0.18
Coamano	2	1	0.05	77	1.9	1.9	903.82
Daphne.Major	18	11	0.34	119	8.0	8.0	1.84
Daphne.Minor	24	0	0.08	93	6.0	12.0	0.34
Darwin	10	7	2.33	168	34.1	290.2	2.85
Eden	8	4	0.03	71	0.4	0.4	17.95
Enderby	2	2	0.18	112	2.6	50.2	0.10

1.1 Plotting

In [22]:
```
# The plot

gala = pd.read_csv("gala.csv")

gala["log_species"] = np.log(gala["Species"])
gala["log_area"] = np.log(gala["Area"])

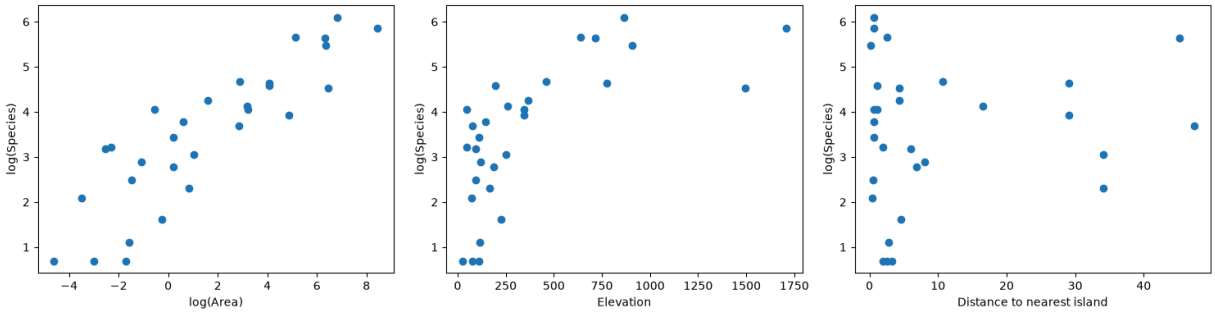
fig, axes = plt.subplots(1, 3, figsize=(15, 4))
```

```
axes[0].scatter(gala["log_area"], gala["log_species"])
axes[0].set_xlabel("log(Area)")
axes[0].set_ylabel("log(Species)")

axes[1].scatter(gala["Elevation"], gala["log_species"])
axes[1].set_xlabel("Elevation")
axes[1].set_ylabel("log(Species)")

axes[2].scatter(gala["Nearest"], gala["log_species"])
axes[2].set_xlabel("Distance to nearest island")
axes[2].set_ylabel("log(Species)")

plt.tight_layout()
plt.show()
```



1.2 Analysis

In [23]:
```
# fit

formula = "Species ~ log_area + Elevation + Nearest + log_area:Elevation + log_area
model = smf.glm(formula, data=gala, family=sm.families.Poisson()).fit()
print(model.summary())

print("explained deviance:", 1 - model.deviance / model.null_deviance)
print("deviance / df:", model.deviance / model.df_resid)
print("Pearson chi2 / df:", model.pearson_chi2 / model.df_resid)
print("AIC:", model.aic)
```

```
                 Generalized Linear Model Regression Results
==============================================================================
Dep. Variable:                Species   No. Observations:                   30
Model:                            GLM   Df Residuals:                       22
Model Family:                 Poisson   Df Model:                            7
Link Function:                    Log   Scale:                          1.0000
Method:                          IRLS   Log-Likelihood:                -232.83
Date:                Mon, 02 Nov 2026   Deviance:                       304.84
Time:                        13:39:56   Pearson chi2:                     305.
No. Iterations:                     6   Pseudo R-squ. (CS):              1.000
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept      3.4404      0.075     46.075      0.000       3.294       3.587
log_area       0.2931      0.025     11.634      0.000       0.244       0.343
Elevation     -0.0004      0.000     -1.154      0.248      -0.001       0.000
Nearest       -0.0227      0.005     -4.950      0.000      -0.032      -0.014
log_area:Elevation     0.0004   4.42e-05      8.866      0.000       0.000       0.000
log_area:Nearest      -0.0034      0.001     -2.272      0.023      -0.006      -0.000
Elevation:Nearest   5.312e-05   1.14e-05      4.657      0.000    3.08e-05    7.55e-05
I(Elevation ** 2)  -1.755e-06   1.73e-07    -10.156      0.000   -2.09e-06   -1.42e-06
==============================================================================
explained deviance: 0.9131694070603483
deviance / df: 13.856302155103656
Pearson chi2 / df: 13.86688608668821
AIC: 481.6696037389449
```

In [24]:
```
# is this a good model?

#Explained deviance is ~0.91, which is decent
#Deviance/df and Pearson chi2/df are high, when they should be about 1, so the mode
#Need a Poisson or negative binomial model
```

1.3 Endemics

In [26]:
```
# The plot
gala["log_end"] = np.log(gala["Endemics"] + 1)
gala["log_area"] = np.log(gala["Area"])

fig, axes = plt.subplots(1, 3, figsize=(15, 4))

axes[0].scatter(gala["log_area"], gala["log_end"])
```

```
axes[0].set_xlabel("log(Area)")
axes[0].set_ylabel("log(Endemics + 1)")

axes[1].scatter(gala["Elevation"], gala["log_end"])
axes[1].set_xlabel("Elevation")
axes[1].set_ylabel("log(Endemics + 1)")

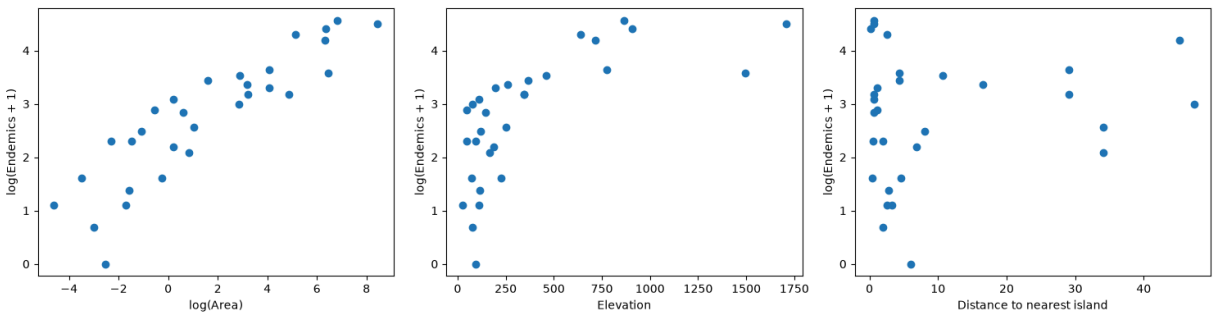
axes[2].scatter(gala["Nearest"], gala["log_end"])
axes[2].set_xlabel("Distance to nearest island")
axes[2].set_ylabel("log(Endemics + 1)")

plt.tight_layout()
plt.show()

# The fit
formula = "Endemics ~ log_area + Elevation + Nearest + log_area:Elevation + log_are
model = smf.glm(formula, data=gala, family=sm.families.Poisson()).fit()
print(model.summary())

print("explained deviance:", 1 - model.deviance / model.null_deviance)
print("deviance / df:", model.deviance / model.df_resid)
print("Pearson chi2 / df:", model.pearson_chi2 / model.df_resid)
print("AIC:", model.aic)

#Explained deviance is ~0.89, which is relatively low and would indicate a good fit
#Deviance/df and Pearson chi2/df are around 3, which is better than the previous mo
#Still would benefit from using Poisson or negative binomial model
```



```
                 Generalized Linear Model Regression Results
==============================================================================
Dep. Variable:                Endemics   No. Observations:                   30
Model:                            GLM   Df Residuals:                       22
Model Family:                 Poisson   Df Model:                            7
Link Function:                    Log   Scale:                          1.0000
Method:                          IRLS   Log-Likelihood:                -104.59
Date:                Mon, 02 Nov 2026   Deviance:                       76.230
Time:                        15:11:39   Pearson chi2:                     69.8
No. Iterations:                     5   Pseudo R-squ. (CS):              1.000
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept      2.3734      0.126     18.831      0.000       2.126       2.620
log_area       0.2564      0.042      6.047      0.000       0.173       0.339
Elevation      0.0005      0.001      0.755      0.451      -0.001       0.002
Nearest       -0.0096      0.007     -1.419      0.156      -0.023       0.004
log_area:Elevation     0.0002   7.38e-05      2.195      0.028    1.73e-05       0.000
log_area:Nearest      -0.0013      0.002     -0.563      0.573      -0.006       0.003
Elevation:Nearest   1.516e-05   1.83e-05      0.831      0.406   -2.06e-05    5.09e-05
I(Elevation ** 2)  -1.119e-06   3.08e-07     -3.632      0.000   -1.72e-06   -5.15e-07
==============================================================================
explained deviance: 0.8974782844351961
deviance / df: 3.464994291384605
Pearson chi2 / df: 3.174868783693108
AIC: 225.18109829043328
```

1.4 Negative binomial model for plant species richness

In [48]:

```python
# estimate the what `alpha` value to use
gala["log_area"] = np.log(gala["Area"])
gala["Elev100"] = gala["Elevation"] / 100   # rescaled so the optimizer behaves

formula = "Species ~ log_area + Elev100 + Nearest + log_area:Elev100 + log_area:Nea

nb_ml = smf.negativebinomial(formula, data=gala).fit(method="bfgs", maxiter=2000)
alpha = nb_ml.params["alpha"]
print("alpha:", alpha)
```

```
Optimization terminated successfully.
         Current function value: 4.503785
         Iterations: 22
         Function evaluations: 33
         Gradient evaluations: 33
alpha: 0.2893474080307853
```

In [49]:
```
# fit Negative binomial
nb_model = smf.glm(formula, data=gala, family=sm.families.NegativeBinomial(alpha=al
print(nb_model.summary())
```

```
                 Generalized Linear Model Regression Results
==============================================================================
Dep. Variable:                 Species   No. Observations:                   30
Model:                             GLM   Df Residuals:                       22
Model Family:        NegativeBinomial   Df Model:                            7
Link Function:                     Log   Scale:                          1.0000
Method:                           IRLS   Log-Likelihood:                -135.11
Date:                 Mon, 02 Nov 2026   Deviance:                       33.434
Time:                         15:26:03   Pearson chi2:                     30.0
No. Iterations:                      10   Pseudo R-squ. (CS):             0.9904
Covariance Type:             nonrobust
==============================================================================
                     coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept            3.5554      0.342     10.406      0.000       2.886       4.225
log_area             0.3894      0.101      3.836      0.000       0.190       0.588
Elev100             -0.1255      0.193     -0.649      0.516      -0.504       0.253
Nearest             -0.0258      0.015     -1.711      0.087      -0.055       0.004
log_area:Elev100     0.0305      0.024      1.250      0.211      -0.017       0.078
log_area:Nearest    -0.0021      0.006     -0.345      0.730      -0.014       0.010
Elev100:Nearest      0.0046      0.005      0.918      0.359      -0.005       0.014
I(Elev100 ** 2)     -0.0115      0.011     -1.048      0.295      -0.033       0.010
==============================================================================
```

In [50]:
```
# is model good?
pois_model = smf.glm(formula, data=gala, family=sm.families.Poisson()).fit()

comp = pd.DataFrame({
    "pois_coef": pois_model.params, "pois_se": pois_model.bse, "pois_p": pois_model
    "nb_coef": nb_model.params, "nb_se": nb_model.bse, "nb_p": nb_model.pvalues,
})
print(comp.round(5))

print("explained deviance:", 1 - nb_model.deviance / nb_model.null_deviance)
print("AIC Poisson:", pois_model.aic, " AIC NB:", nb_ml.aic)
```

```
                  pois_coef  pois_se   pois_p  nb_coef    nb_se     nb_p
Intercept           3.44038  0.07467  0.00000  3.55543  0.34168  0.00000
log_area            0.29313  0.02520  0.00000  0.38936  0.10149  0.00012
Elev100            -0.04042  0.03502  0.24842 -0.12547  0.19320  0.51606
Nearest            -0.02266  0.00458  0.00000 -0.02580  0.01508  0.08709
log_area:Elev100    0.03916  0.00442  0.00000  0.03054  0.02443  0.21124
log_area:Nearest   -0.00336  0.00148  0.02306 -0.00209  0.00606  0.73021
Elev100:Nearest     0.00531  0.00114  0.00000  0.00456  0.00496  0.35858
I(Elev100 ** 2)    -0.01755  0.00173  0.00000 -0.01145  0.01093  0.29482
explained deviance: 0.8065740520907907
AIC Poisson: 481.6696037389439  AIC NB: 288.22712944286684
```

In [51]:
```
# dispersion?
print("Poisson Pearson/df:", pois_model.pearson_chi2 / pois_model.df_resid)
print("NB Pearson/df:", nb_model.pearson_chi2 / nb_model.df_resid)
print("NB deviance/df:", nb_model.deviance / nb_model.df_resid)
#NB Pearson/df and NB deviance/df are close to 1, which mean this model is a better
```

```
Poisson Pearson/df: 13.866886087495756
NB Pearson/df: 1.3631119959860518
NB deviance/df: 1.5197187025856478
```

1.5 Negative binomial model for endemic plant species richness

In [52]:
```
# what `alpha` value to use?
gala["Elev100"] = gala["Elevation"] / 100   # same rescaling as in step 4 (safe to

formula_end = "Endemics ~ log_area + Elev100 + Nearest + log_area:Elev100 + log_are

nb_ml_end = smf.negativebinomial(formula_end, data=gala).fit(method="bfgs", maxiter
alpha_end = nb_ml_end.params["alpha"]
print("alpha:", alpha_end)
```

```
Optimization terminated successfully.
         Current function value: 3.318762
         Iterations: 25
         Function evaluations: 43
         Gradient evaluations: 36
alpha: 0.07653334453425853
```

In [53]:
```
# Negative binomial fit
nb_end = smf.glm(formula_end, data=gala, family=sm.families.NegativeBinomial(alpha=
print(nb_end.summary())
```

```
                 Generalized Linear Model Regression Results
==============================================================================
Dep. Variable:                Endemics   No. Observations:                   30
Model:                            GLM   Df Residuals:                       22
Model Family:        NegativeBinomial   Df Model:                            7
Link Function:                    Log   Scale:                          1.0000
Method:                          IRLS   Log-Likelihood:                -99.563
Date:                Mon, 02 Nov 2026   Deviance:                       38.986
Time:                        15:26:50   Pearson chi2:                     33.0
No. Iterations:                     9   Pseudo R-squ. (CS):             0.9995
Covariance Type:            nonrobust
==============================================================================
                     coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept            2.4594      0.216     11.369      0.000       2.035       2.883
log_area             0.3250      0.069      4.683      0.000       0.189       0.461
Elev100             -0.0074      0.119     -0.062      0.950      -0.240       0.225
Nearest             -0.0144      0.010     -1.398      0.162      -0.034       0.006
log_area:Elev100     0.0088      0.015      0.585      0.558      -0.021       0.038
log_area:Nearest    -0.0009      0.004     -0.220      0.826      -0.009       0.007
Elev100:Nearest      0.0017      0.003      0.548      0.584      -0.004       0.008
I(Elev100 ** 2)     -0.0067      0.007     -1.031      0.303      -0.020       0.006
==============================================================================
```

In [54]:

```
# is model good?
pois_end = smf.glm(formula_end, data=gala, family=sm.families.Poisson()).fit()

comp_end = pd.DataFrame({
    "pois_coef": pois_end.params, "pois_se": pois_end.bse, "pois_p": pois_end.pvalu
    "nb_coef": nb_end.params, "nb_se": nb_end.bse, "nb_p": nb_end.pvalues,
})
print(comp_end.round(5))

print("explained deviance:", 1 - nb_end.deviance / nb_end.null_deviance)
print("AIC Poisson:", pois_end.aic, " AIC NB:", nb_ml_end.aic)

print("Poisson Pearson/df:", pois_end.pearson_chi2 / pois_end.df_resid)
print("NB Pearson/df:", nb_end.pearson_chi2 / nb_end.df_resid)
print("NB deviance/df:", nb_end.deviance / nb_end.df_resid)
#NB Pearson/df and NB deviance/df are close to 1, which mean this model is a better
```

```
                pois_coef  pois_se   pois_p  nb_coef    nb_se     nb_p
Intercept         2.37338  0.12604  0.00000  2.45943  0.21633  0.00000
log_area          0.25638  0.04239  0.00000  0.32503  0.06940  0.00000
Elev100           0.04607  0.06106  0.45055 -0.00737  0.11854  0.95039
Nearest          -0.00960  0.00677  0.15596 -0.01436  0.01027  0.16226
log_area:Elev100  0.01621  0.00738  0.02818  0.00876  0.01496  0.55825
log_area:Nearest -0.00134  0.00238  0.57336 -0.00088  0.00400  0.82590
Elev100:Nearest   0.00152  0.00183  0.40617  0.00173  0.00315  0.58401
I(Elev100 ** 2)  -0.01119  0.00308  0.00028 -0.00675  0.00655  0.30268
explained deviance: 0.8538961797838361
AIC Poisson: 225.18109829043345  AIC NB: 217.12572414681398
Poisson Pearson/df: 3.174868783743285
NB Pearson/df: 1.4994979426586865
NB deviance/df: 1.7720792482963885
```

1.6 Simplify models

In [55]:
```
# To ignore all warnings
warnings.filterwarnings('ignore')

from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler

# Perform stepwise selection with r2 criterion

## Model for Species
gala["log_area"] = np.log(gala["Area"])

X = gala[["log_area", "Elevation", "Nearest"]].copy()
X["log_area:Elevation"] = X["log_area"] * X["Elevation"]
X["log_area:Nearest"] = X["log_area"] * X["Nearest"]
X["Elevation:Nearest"] = X["Elevation"] * X["Nearest"]
X["Elevation^2"] = X["Elevation"] ** 2

model = make_pipeline(StandardScaler(), PoissonRegressor(alpha=0, max_iter=1000))

sfs_species = SequentialFeatureSelector(model, k_features="best", forward=True,
                                        floating=True, scoring="r2", cv=5)
sfs_species = sfs_species.fit(X, gala["Species"])
print("selected:", sfs_species.k_feature_names_)
print("r2:", sfs_species.k_score_)
```

```
selected: ('log_area', 'log_area:Elevation', 'Elevation^2')
r2: 0.7483460035603817
```

In [56]:
```
# To ignore all warnings
warnings.filterwarnings('ignore')

# Perform stepwise selection with r2 criterion

## Model for Endemics
sfs_endemics = SequentialFeatureSelector(model, k_features="best", forward=True,
                                         floating=True, scoring="r2", cv=5)
sfs_endemics = sfs_endemics.fit(X, gala["Endemics"])
print("selected:", sfs_endemics.k_feature_names_)
print("r2:", sfs_endemics.k_score_)
```

```
selected: ('log_area', 'log_area:Elevation', 'Elevation^2')
r2: 0.7431728807541607
```

1.7 Plot data and regression lines with confidence intervals

Note: In python, when are ploting against a log-transformed "X", you have to force the x-axis to be log-scaled by adding `plt.xscale('log')` to the plotting code.

In [68]:
```
# Create the scatter plot
# Color palette
ocean_blue = "#00A6A6"
hibiscus_pink = "#E85D8E"
```

```
mango_orange = "#FF9F43"
deep_teal = "#006D77"
lime_green = "#A8C94A"

fig, axes = plt.subplots(2, 1, figsize=(5, 5))

# Total species richness
axes[0].scatter(
    gala["Area"], gala["Species"],
    color=hibiscus_pink, edgecolor="white",
    linewidth=0.6, s=45, alpha=0.85,
    label="Observed data"
)
axes[0].plot(
    area_range, pred_species["mean"],
    color=ocean_blue, linewidth=2.5,
    label="Negative binomial model"
)
axes[0].fill_between(
    area_range,
    pred_species["mean_ci_lower"],
    pred_species["mean_ci_upper"],
    color=ocean_blue, alpha=0.2,
    label="95% confidence interval"
)

axes[0].set_xscale("log")
axes[0].set_xlabel("Area (log scale)")
axes[0].set_ylabel("Number of plant species")
axes[0].set_title("Total species richness", color=deep_teal)
axes[0].legend(frameon=False)

# Endemic species richness
axes[1].scatter(
    gala["Area"], gala["Endemics"],
    color=mango_orange, edgecolor="white",
    linewidth=0.6, s=45, alpha=0.85,
    label="Observed data"
)
axes[1].plot(
    area_range, pred_endemics["mean"],
    color=deep_teal, linewidth=2.5,
    label="Negative binomial model"
)
axes[1].fill_between(
    area_range,
    pred_endemics["mean_ci_lower"],
    pred_endemics["mean_ci_upper"],
    color=lime_green, alpha=0.22,
    label="95% confidence interval"
)

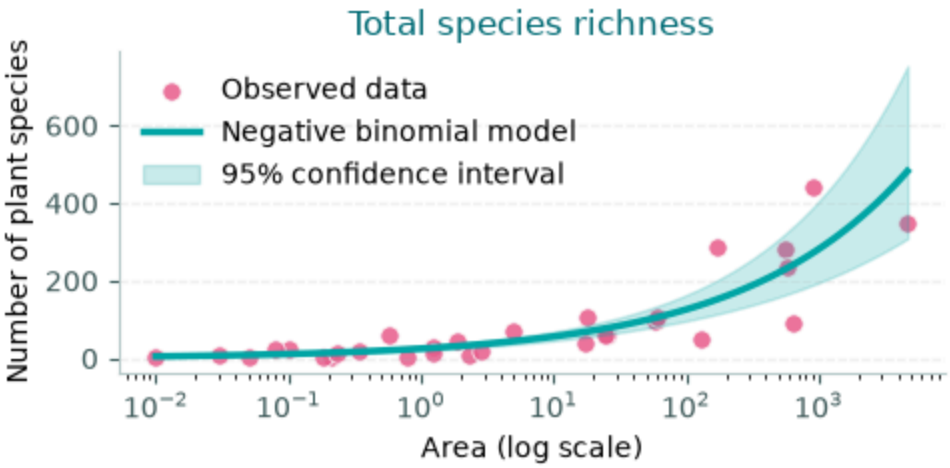
axes[1].set_xscale("log")
axes[1].set_xlabel("Area (log scale)")
axes[1].set_ylabel("Number of endemic species")
axes[1].set_title("Endemic species richness", color=deep_teal)
```

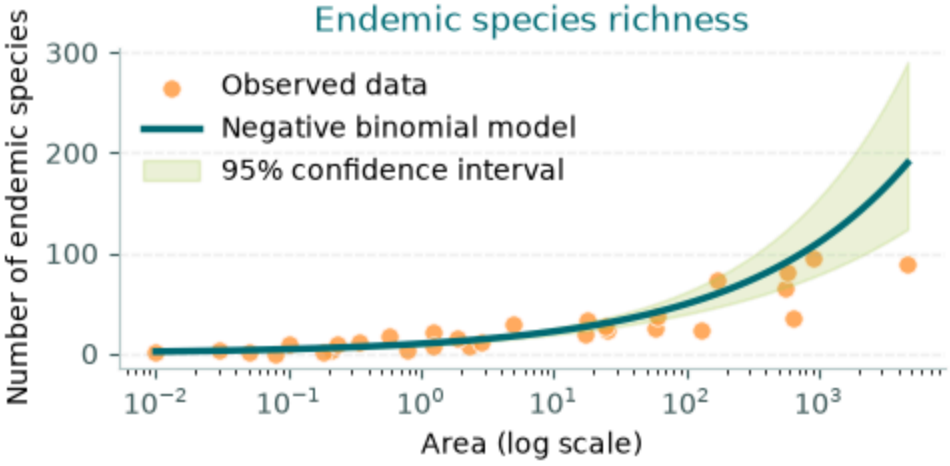
```
axes[1].legend(frameon=False)

# Clean, tropical styling
for ax in axes:
    ax.spines["top"].set_visible(False)
    ax.spines["right"].set_visible(False)
    ax.spines["left"].set_color("#B7C9C9")
    ax.spines["bottom"].set_color("#B7C9C9")
    ax.tick_params(colors="#405B5B")
    ax.grid(axis="y", alpha=0.18, linestyle="--")

fig.patch.set_facecolor("#FFFFFF")
for ax in axes:
    ax.set_facecolor("#FFFFFF")

plt.tight_layout()
plt.show()
```





1.8 Interpretation

In [69]:
```
print(species_bic.summary())
print(endemics_bic.summary())
```

```
                 Generalized Linear Model Regression Results
==============================================================================
Dep. Variable:                Species   No. Observations:                   30
Model:                            GLM   Df Residuals:                       28
Model Family:        NegativeBinomial   Df Model:                            1
Link Function:                    Log   Scale:                          1.0000
Method:                          IRLS   Log-Likelihood:                -139.51
Date:                Mon, 02 Nov 2026   Deviance:                       42.223
Time:                        15:49:58   Pearson chi2:                     39.4
No. Iterations:                     6   Pseudo R-squ. (CS):             0.9871
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept      3.2258      0.120     26.871      0.000       2.990       3.461
log_area       0.3495      0.031     11.316      0.000       0.289       0.410
==============================================================================
                 Generalized Linear Model Regression Results
==============================================================================
Dep. Variable:                Endemics   No. Observations:                   30
Model:                            GLM   Df Residuals:                       26
Model Family:        NegativeBinomial   Df Model:                            3
Link Function:                    Log   Scale:                          1.0000
Method:                          IRLS   Log-Likelihood:                -100.27
Date:                Mon, 02 Nov 2026   Deviance:                       40.394
Time:                        15:49:58   Pearson chi2:                     34.3
No. Iterations:                     7   Pseudo R-squ. (CS):             0.9995
Covariance Type:            nonrobust
================================================================================
                   coef    std err          z      P>|z|      [0.025      0.975]
--------------------------------------------------------------------------------
Intercept        2.4446      0.104     23.454      0.000       2.240       2.649
log_area         0.3503      0.031     11.278      0.000       0.289       0.411
Nearest         -0.0111      0.005     -2.232      0.026      -0.021      -0.001
I(Elev100 ** 2) -0.0034      0.001     -2.613      0.009      -0.006      -0.001
================================================================================
```

In [ ]:
```
#Interpretation
/// In the full negative binomial models, both coefficients for species richness an
///indicating that island area is positively associated with both total and endemic
///The models also suggest that island area is more important than other geographic
///species richness///
```